Non-Functional Requirements & SLOs

Performance, resiliency, observability and operability targets

Document status: CURRENT unless otherwise identified | Enterprise delivery assurance training corpus

1. Purpose

Northstar Commerce is preparing a multi-release modernization of its digital commerce platform. This specification is intentionally detailed and includes overlapping business, API, testing, and governance content so that delivery-assurance systems must reason across multiple artifacts rather than rely on one isolated paragraph.

Non-functional requirements must be measurable. Statements such as fast, scalable, resilient, or user friendly are not sufficient test oracles without thresholds or observable criteria.

1.1 Operational Guidance

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2. Performance Targets

Performance tests should use representative traffic mix, realistic payloads, and clearly defined percentile measurements. Averages alone are not sufficient for release gates.

NFR	Target	Measurement
Checkout API p95 latency	<= 800 ms	Load test under representative mix
Payment status p95 latency	<= 500 ms	Service test excluding client network
Availability	99.95% monthly	Service telemetry
Recovery	No duplicate business transaction after retry	Failure injection + state reconciliation
Audit correlation	>= 99.9% events carry correlation ID	Telemetry query

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3. Resiliency

Retries must be bounded and should not create duplicate business transactions. Dependency timeouts should produce observable telemetry and a recoverable business state.

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4. Observability

Critical workflows should carry a correlation identifier across service boundaries. Operational evidence should allow a failed test to be related to relevant logs or traces.

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5. Capacity and Degradation

Under overload, the platform should degrade predictably and protect critical dependencies. Test evidence should distinguish hard failure from controlled rejection.

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